

Project Design Phase

Problem – Solution Fit

Date: 12-07-2026

Team ID: SWTID-2026-5429

Project Name: UCab – MERN Stack Cab Booking System

Problem–Solution Fit

UCab – MERN Stack Cab Booking System			
Problem – Solution Fit Canvas			
Date: 15 February 2025		Team ID: UCAB_Team01	Project Name: UCab – MERN Stack Cab Booking System
		Maximum Marks: 2 Marks	
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who are your customers? - Daily commuters - College students - Office employees - Tourists and travelers - People needing quick transportation - Future: Taxi drivers and fleet operators	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices? - Limited internet connectivity - Budget constraints - Peak-hour cab availability - Lack of transparent fare information - Limited technical knowledge for some users	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem? - Local taxi services - Auto-rickshaws - Public transportation - Phone-based cab booking - Existing ride-booking applications Limitations: - Longer waiting time - No booking history - Manual booking process - Inconsistent pricing - Poor user experience
	2. JOBS-TO-BE-DONE / PROBLEMS What jobs/problems are customers trying to solve? - Book a cab quickly and easily. - Find available cabs without making phone calls. - Know the estimated fare before booking. - Save time while booking rides. - View previous booking history. - Access transportation through a simple web application.	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? - Traditional booking methods are slow. - No centralized booking platform. - Difficulty in managing ride information. - Lack of secure authentication. - Poor communication between users and service providers. - Limited transparency in fare estimation.	7. BEHAVIOUR How does your customer do to address the problem and get the job done? - Search online for nearby transportation. - Compare different ride options. - Use mobile or web applications. - Choose services based on convenience and fare. - Prefer fast booking and instant confirmation.
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? i.e. seeing their neighbor installing solar panels, reading about a more efficient solution in the news. - Need to reach office or college on time. - Traveling to airports, railway stations, or bus stations. - Public transport is unavailable. - Bad weather conditions. - Emergency travel requirements. - Visiting unfamiliar locations.	10. YOUR SOLUTION If you are solving the problem/ painpoints, write down your current solution here. UCab is a MERN Stack-based Full Stack Cab Booking Web Application that provides: - Secure User Registration & Login - JWT Authentication - Cab Booking System - Pickup & Destination Selection - Fare Estimation - Booking Confirmation - Booking History - User Profile Management - Responsive User Interface - MongoDB Database Integration Future Enhancements - Google Maps Integration - Live GPS Tracking - Driver Dashboard - Admin Dashboard - Ride Rating System - Push Notifications - AI-based Fare Prediction	8. CHANNELS OF BEHAVIOUR 8.1 ONLINE - UCab Web Application - Google Search - Social Media - GitHub Repository - College Project Demonstrations 8.2 OFFLINE - Word-of-mouth recommendations - College exhibitions - Technical seminars - Workshops - Local transportation services
	4. EMOTIONS: BEFORE / AFTER Before Using UCab - Stress about finding transportation. - Uncertainty regarding ride availability. - Frustration due to waiting. - Confusion about fare. After Using UCab - Convenient booking experience. - Confidence in fare estimation. - Saves time. - Comfortable and hassle-free booking. - Higher customer satisfaction.		

Summary

Problem

Manual and time-consuming cab booking

Difficulty finding transportation

Lack of fare transparency

UCab Solution

Online cab booking through a responsive web application

Simple ride booking interface

Estimated fare calculation before booking

Problem**UCab Solution**

No centralized ride management

Booking history and profile management

Insecure user authentication

JWT authentication with encrypted passwords

Poor user experience

Responsive and intuitive React-based interface

Limited scalability

Modular MERN architecture supporting future enhancements